PolicyGraphAudit-RT

Compliance Audit Report

Heterogeneous GNN Privacy Analysis

[redacted] HD TV: [redacted]

limehd.ru.lite

Developer	[redacted]
Genre	ENTERTAINMENT
Installs	5,000,000+
Audit Date	2026-04-26

Overall Risk Score **0.74** **HIGH**

0 Consistent	**7** Mismatches	**0** Over-disclosure	**1** Undeclared

Research prototype · Ongoing research | Policy: http://images-iptv2021.cdnvideo.ru/docs/instructions/privacy...

Executive Summary

[redacted] HD TV: [redacted] declares 7 data type(s) collected and 0 shared in its Play Data Safety label, across 8 data type(s) analyzed. Our heterogeneous GNN audit model identifies **1 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **7 policy-vs-label mismatch(es)**. Overall risk score: **0.74 (HIGH)**.

Top discrepancies by risk:

- **diagnostics**: Undeclared Collection (confidence 0.99)
- **app_interactions**: Policy Label Mismatch (confidence 1.00)
- **device_id**: Policy Label Mismatch (confidence 1.00)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
diagnostics	**Undeclared Collection**	0.99	label:N policy:N sdk:Y (1, 2)
app_interactions	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:N (1, 2)
device_id	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:Y (1, 2)
installed_apps	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:Y (1, 2)
crash_logs	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:Y (1, 2)
user_ids	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:N (1, 2)
precise_location	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:N (1, 2)
approximate_location	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:N (1, 2)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: diagnostics — Undeclared Collection (confidence 0.99)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2

Recommended action: Add 'diagnostics' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: app_interactions — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2

Recommended action: Update the privacy policy to disclose 'app_interactions' collection, aligning it with the Play Data Safety label.

Discrepancy: device_id — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2

Recommended action: Update the privacy policy to disclose 'device_id' collection, aligning it with the Play Data Safety label.

Discrepancy: installed_apps — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2

Recommended action: Update the privacy policy to disclose 'installed_apps' collection, aligning it with the Play Data Safety label.

Discrepancy: crash_logs — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2

Recommended action: Update the privacy policy to disclose 'crash_logs' collection, aligning it with the Play Data Safety label.

Discrepancy: user_ids — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2

Recommended action: Update the privacy policy to disclose 'user_ids' collection, aligning it with the Play Data Safety label.

Discrepancy: precise_location — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2

Recommended action: Update the privacy policy to disclose 'precise_location' collection, aligning it with the Play Data Safety label.

Discrepancy: approximate_location — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2

Recommended action: Update the privacy policy to disclose 'approximate_location' collection, aligning it with the Play Data Safety label.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer